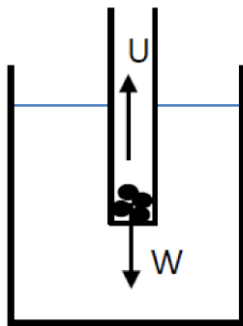


4 (a) (i)



W: Weight of loaded test-tube

or

W_L : weight of lead shots and W_T : weight of test-tube

U: Upthrust / Force by fluid on loaded test-tube

Legend for both W and U

B1

W and U of same length and act along the same vertical line

B1

(Note that arrow of W should originate from C.G. of loaded test-tube while arrow of U should originate from centre of mass of the displaced fluid.)

(ii) Summing forces vertically: $mg = \text{Upthrust} = \rho ALg$

M1

mass of loaded test tube, $m = \rho AL$

A1

(b) (i) Resultant force (in vector notation): $F = \text{Upthrust} + \text{Weight}$

$$F = -\rho A(L+x)g + mg = -\rho ALg + mg - \rho A x g$$

M1

Since $mg + (-\rho ALg) = 0$

M1

$$F = -\rho A x g$$

A0

(ii) $F = -\rho A x g = ma$

$$a = -\left(\frac{\rho A g}{m}\right)x$$

M1

$$= -\left(\frac{\rho A g}{\rho A L}\right)x$$

M1

Since loaded test-tube is in SHM: $a = -\omega^2 x$

M1

$$\omega^2 = \frac{g}{L}, \text{ so } \omega = \sqrt{\frac{g}{L}}$$

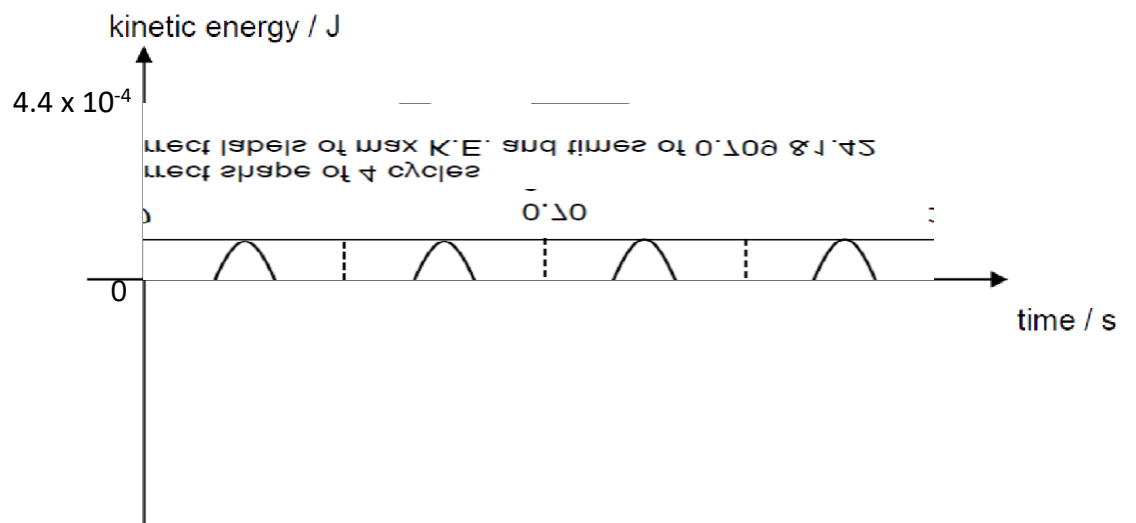
(iii) Total energy $= \frac{1}{2} m \omega^2 x_0^2 = \frac{1}{2} (0.050)(9.81/0.125)(0.015)^2$

M1

$$= 0.00044 \text{ J}$$

A1

(iv)



correct shape ($\sin^2$ not modulus) of 4 "humps" with k.e. starting from zero

B1

correct label of max K.E.

B1

5 (c) (i) Moves from the top elite overlap and supersedes (debrief point) at points on the